

Battery_Optimizer

May 3, 2026

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[6]: import pandas as pd
import gurobipy as gp
from gurobipy import GRB

# Load in Account A data
df = pd.read_csv("account_a.csv")

# Convert Datetime variable so it is parsed correctly
df["Datetime"] = pd.to_datetime(df["Datetime"], format="mixed", errors="coerce")

# Extract date and hour
df["date_only"] = df["Datetime"].dt.date
df["hour"] = df["Datetime"].dt.hour

# First 2 weeks (14 days) only
time = sorted(df["date_only"].unique())[:14]
df_14 = df[df["date_only"].isin(time)].copy()

# Map days and hours indexes
day_map = {day: i + 1 for i, day in enumerate(time)}
df_14["d"] = df_14["date_only"].map(day_map)
df_14["t"] = df_14["hour"]

# Calculate Price and Quantity
hourly = (
    df_14.groupby(["d", "t"], as_index=False)
    .agg({
        "Quantity": "sum",
        "Settlement.Point.Price.KWH": "mean"
    })
)

Price = {(row["d"], row["t"]): row["Settlement.Point.Price.KWH"] for _, row in
    hourly.iterrows()}
Quantity = {(row["d"], row["t"]): row["Quantity"] for _, row in
    hourly.iterrows()}
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# Create set indexes
DT = sorted(list(Quantity.keys()))

# Max batteries for 2 week period
B_max = 30

# Battery cost
C_battery = 14900

# Charging/Discharging rate per battery per hour
r_ch = 5
r_dis = 11.5

m = gp.Model("battery_model_account_a")

# Decision Variables
x_A = m.addVar(vtype=GRB.INTEGER, lb=0, ub=B_max)
e = m.addVars(range(1, B_max + 1), vtype=GRB.BINARY)
y = m.addVars(range(1, B_max + 1), DT, vtype=GRB.BINARY)
charge = m.addVars(DT, vtype=GRB.CONTINUOUS, lb=0)
discharge = m.addVars(DT, vtype=GRB.CONTINUOUS, lb=0)
cs = m.addVars(DT, vtype=GRB.CONTINUOUS, lb=0)
charge_b = m.addVars(range(1, B_max+1), DT, lb=0, vtype=GRB.CONTINUOUS)
discharge_b = m.addVars(range(1, B_max+1), DT, lb=0, vtype=GRB.CONTINUOUS)

# Constraints
# sum_b e_A,b = x_A
m.addConstr(
    gp.quicksum(e[b] for b in range(1, B_max + 1)) == x_A,
    name="num_active_batteries"
)

# y_b,A,t,d <= e_A,b
for b in range(1, B_max + 1):
    for d, t in DT:
        m.addConstr(y[b, d, t] <= e[b])

# g_{A,t,d} = Quantity_{A,t,d}
for d, t in DT:
    m.addConstr(discharge[d, t] == Quantity[d, t])

# charging state capacity constraints
for d, t in DT:
    m.addConstr(cs[d, t] <= 13.5 * x_A)
    m.addConstr(cs[d, t] >= 2.7 * x_A)

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# charging/discharging rate limits
for d, t in DT:
    m.addConstr(charge[d, t] <= r_ch * x_A)
    m.addConstr(discharge[d, t] <= r_dis * x_A)

# Charging state across time
DT_sorted = sorted(DT)

# Charging state constraints
for i, (d, t) in enumerate(DT_sorted):
    if i == 0:
        m.addConstr(
            cs[d, t] == 13.5 * x_A + charge[d, t] - discharge[d, t]
        )
    else:
        prev_d, prev_t = DT_sorted[i - 1]
        m.addConstr(
            cs[d, t] == cs[prev_d, prev_t] + charge[d, t] - discharge[d, t]
        )

# Battery charging constraint
for b in range(1, B_max+1):
    for d, t in DT:
        m.addConstr(
            charge_b[b, d, t] <= r_ch * y[b, d, t],
            name=f"charge_mode_b{b}_d{d}_t{t}"
        )

# Battery discharging constraint
for b in range(1, B_max+1):
    for d, t in DT:
        m.addConstr(
            discharge_b[b, d, t] <= r_dis * (e[b] - y[b, d, t]),
            name=f"discharge_mode_b{b}_d{d}_t{t}"
        )

# Sum of all charge and discharge for each battery must equal the totals
for d, t in DT:
    m.addConstr(
        charge[d, t] == gp.quicksum(charge_b[b, d, t] for b in range(1,
↪B_max+1)),
        name=f"charge_total_d{d}_t{t}"
    )

    m.addConstr(

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        discharge[d, t] == gp.quicksum(discharge_b[b, d, t] for b in range(1,
↪B_max+1)),
        name=f"discharge_total_d{d}_t{t}"
    )

# Objective Function
energy_cost = gp.quicksum(
    charge[d, t] * Price[d, t]
    for d, t in DT
)

battery_cost = x_A * C_battery

m.setObjective(energy_cost + battery_cost, GRB.MINIMIZE)

# Optimize
m.optimize()

# Print results
if m.status == GRB.OPTIMAL:
    print("\nOptimal Solution")
    print(f"x_A = {x_A.X}")
    print(f"Energy cost = {energy_cost.getValue():.4f}")
    print(f"Battery cost = {battery_cost.getValue():.4f}")
    print(f"Total Cost = {m.objVal:.4f}")

    # Display charge/discharge values for some iterations just to check
    print("\nSample hourly values:")
    for d, t in DT_sorted[:600]:
        print(
            f"(d={d}, t={t}) | charge = {charge[d,t].X:.4f} kWh, "
            f"discharge = {discharge[d,t].X:.4f} kWh, charge_state = {cs[d,t].X:
↪.4f} kWh, price = ${Price[(d,t)]:.4f}"
        )
    else:
        print("No optimal solution found.")

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Gurobi Optimizer version 12.0.0 build v12.0.0rc1 (win64 - Windows 11+.0 (26200.2))

CPU model: Intel(R) Core(TM) Ultra 7 155U, instruction set [SSE2|AVX|AVX2]
 Thread count: 12 physical cores, 14 logical processors, using up to 14 threads

Optimize a model with 22639 rows, 21514 columns and 65866 nonzeros
 Model fingerprint: 0x4a8a9661
 Variable types: 14553 continuous, 6961 integer (6960 binary)

Coefficient statistics:

Matrix range [1e+00, 1e+01]
 Objective range [6e-03, 1e+04]
 Bounds range [1e+00, 3e+01]
 RHS range [1e+00, 5e+00]

Presolve removed 696 rows and 462 columns

Presolve time: 0.22s

Presolved: 21943 rows, 21052 columns, 64473 nonzeros

Variable types: 14091 continuous, 6961 integer (6960 binary)

Found heuristic solution: objective 447017.08350

Found heuristic solution: objective 447016.33947

Root relaxation: objective 2.272915e+04, 18628 iterations, 1.70 seconds (0.91 work units)

Nodes		Current Node			Objective Bounds			Work	
Expl	Unexpl	Obj	Depth	IntInf	Incumbent	BestBd	Gap	It/Node	Time
	0	0	22729.1491	0	225 447016.339	22729.1491	94.9%	-	3s
H	0	0			29826.710757	22729.1491	23.8%	-	3s
H	0	0			29825.977967	22729.1491	23.8%	-	3s
	0	0	cutoff	0	29825.9780	29825.9780	0.00%	-	4s

Cutting planes:

Implied bound: 2
 MIR: 7
 Flow cover: 61
 RLT: 1
 Relax-and-lift: 10

Explored 1 nodes (33038 simplex iterations) in 4.91 seconds (2.74 work units)
 Thread count was 14 (of 14 available processors)

Solution count 4: 29826 29826.7 447016 447017

Optimal solution found (tolerance 1.00e-04)

Best objective 2.982597796720e+04, best bound 2.982597796720e+04, gap 0.0000%

Optimal Solution

x_A = 2.0
 Energy cost = 25.9780
 Battery cost = 29,800.0000
 Total Cost = 29,825.9780

Sample hourly values:

(d=1.0, t=0.0) | charge = 0.0000 kWh, discharge = 1.2592 kWh, charge_state = 25.7407 kWh, price = \$0.0613
 (d=1.0, t=1.0) | charge = 0.0000 kWh, discharge = 1.2765 kWh, charge_state =

24.4642 kWh, price = \$0.0577
 (d=1.0, t=2.0) | charge = 0.0000 kWh, discharge = 1.2592 kWh, charge_state =
 23.2050 kWh, price = \$0.0536
 (d=1.0, t=3.0) | charge = 1.0375 kWh, discharge = 0.9832 kWh, charge_state =
 23.2593 kWh, price = \$0.0500
 (d=1.0, t=4.0) | charge = 5.0000 kWh, discharge = 1.2592 kWh, charge_state =
 27.0000 kWh, price = \$0.0489
 (d=1.0, t=5.0) | charge = 0.9660 kWh, discharge = 0.9660 kWh, charge_state =
 27.0000 kWh, price = \$0.0542
 (d=1.0, t=6.0) | charge = 0.0000 kWh, discharge = 1.2592 kWh, charge_state =
 25.7408 kWh, price = \$0.0607
 (d=1.0, t=7.0) | charge = 0.0000 kWh, discharge = 1.2765 kWh, charge_state =
 24.4643 kWh, price = \$0.0622
 (d=1.0, t=8.0) | charge = 4.5023 kWh, discharge = 1.9665 kWh, charge_state =
 27.0000 kWh, price = \$0.0574
 (d=1.0, t=9.0) | charge = 2.2598 kWh, discharge = 2.2597 kWh, charge_state =
 27.0000 kWh, price = \$0.0613
 (d=1.0, t=10.0) | charge = 2.6910 kWh, discharge = 2.6910 kWh, charge_state =
 27.0000 kWh, price = \$0.0630
 (d=1.0, t=11.0) | charge = 0.0000 kWh, discharge = 3.5362 kWh, charge_state =
 23.4638 kWh, price = \$0.0711
 (d=1.0, t=12.0) | charge = 0.0000 kWh, discharge = 3.6570 kWh, charge_state =
 19.8068 kWh, price = \$0.0739
 (d=1.0, t=13.0) | charge = 0.0000 kWh, discharge = 4.0020 kWh, charge_state =
 15.8048 kWh, price = \$0.0826
 (d=1.0, t=14.0) | charge = 0.0000 kWh, discharge = 4.2953 kWh, charge_state =
 11.5095 kWh, price = \$0.0999
 (d=1.0, t=21.0) | charge = 0.0000 kWh, discharge = 2.0355 kWh, charge_state =
 9.4740 kWh, price = \$0.0936
 (d=1.0, t=22.0) | charge = 0.0000 kWh, discharge = 2.2597 kWh, charge_state =
 7.2143 kWh, price = \$0.0781
 (d=1.0, t=23.0) | charge = 2.1195 kWh, discharge = 2.1562 kWh, charge_state =
 7.1775 kWh, price = \$0.0647
 (d=2.0, t=0.0) | charge = 5.0000 kWh, discharge = 1.8802 kWh, charge_state =
 10.2973 kWh, price = \$0.0640
 (d=2.0, t=1.0) | charge = 5.0000 kWh, discharge = 1.7595 kWh, charge_state =
 13.5378 kWh, price = \$0.0594
 (d=2.0, t=2.0) | charge = 5.0000 kWh, discharge = 1.6732 kWh, charge_state =
 16.8645 kWh, price = \$0.0499
 (d=2.0, t=3.0) | charge = 5.0000 kWh, discharge = 1.7250 kWh, charge_state =
 20.1395 kWh, price = \$0.0443
 (d=2.0, t=4.0) | charge = 5.0000 kWh, discharge = 1.6732 kWh, charge_state =
 23.4663 kWh, price = \$0.0502
 (d=2.0, t=5.0) | charge = 5.0000 kWh, discharge = 1.4663 kWh, charge_state =
 27.0000 kWh, price = \$0.0565
 (d=2.0, t=6.0) | charge = 0.0000 kWh, discharge = 1.5697 kWh, charge_state =
 25.4303 kWh, price = \$0.0720
 (d=2.0, t=7.0) | charge = 0.0000 kWh, discharge = 1.6042 kWh, charge_state =

23.8260 kWh, price = \$0.0712
 (d=2.0, t=8.0) | charge = 5.0000 kWh, discharge = 2.6738 kWh, charge_state =
 26.1523 kWh, price = \$0.0628
 (d=2.0, t=9.0) | charge = 0.6722 kWh, discharge = 3.4155 kWh, charge_state =
 23.4090 kWh, price = \$0.0652
 (d=2.0, t=10.0) | charge = 0.0000 kWh, discharge = 3.9675 kWh, charge_state =
 19.4415 kWh, price = \$0.0771
 (d=2.0, t=11.0) | charge = 0.0000 kWh, discharge = 4.5540 kWh, charge_state =
 14.8875 kWh, price = \$0.0947
 (d=2.0, t=20.0) | charge = 0.0000 kWh, discharge = 3.0015 kWh, charge_state =
 11.8860 kWh, price = \$0.1006
 (d=2.0, t=21.0) | charge = 0.0000 kWh, discharge = 2.2770 kWh, charge_state =
 9.6090 kWh, price = \$0.0835
 (d=2.0, t=22.0) | charge = 0.0000 kWh, discharge = 2.3632 kWh, charge_state =
 7.2457 kWh, price = \$0.0765
 (d=2.0, t=23.0) | charge = 0.0000 kWh, discharge = 1.8458 kWh, charge_state =
 5.4000 kWh, price = \$0.0706
 (d=3.0, t=0.0) | charge = 2.1218 kWh, discharge = 2.1218 kWh, charge_state =
 5.4000 kWh, price = \$0.0603
 (d=3.0, t=1.0) | charge = 4.5030 kWh, discharge = 1.8458 kWh, charge_state =
 8.0573 kWh, price = \$0.0507
 (d=3.0, t=2.0) | charge = 5.0000 kWh, discharge = 1.7423 kWh, charge_state =
 11.3150 kWh, price = \$0.0366
 (d=3.0, t=3.0) | charge = 5.0000 kWh, discharge = 1.7423 kWh, charge_state =
 14.5728 kWh, price = \$0.0301
 (d=3.0, t=4.0) | charge = 5.0000 kWh, discharge = 1.7423 kWh, charge_state =
 17.8305 kWh, price = \$0.0274
 (d=3.0, t=5.0) | charge = 5.0000 kWh, discharge = 2.2252 kWh, charge_state =
 20.6053 kWh, price = \$0.0319
 (d=3.0, t=6.0) | charge = 5.0000 kWh, discharge = 1.5525 kWh, charge_state =
 24.0528 kWh, price = \$0.0424
 (d=3.0, t=7.0) | charge = 5.0000 kWh, discharge = 2.0528 kWh, charge_state =
 27.0000 kWh, price = \$0.0449
 (d=3.0, t=8.0) | charge = 2.3775 kWh, discharge = 2.8635 kWh, charge_state =
 26.5140 kWh, price = \$0.0508
 (d=3.0, t=9.0) | charge = 0.0000 kWh, discharge = 3.4845 kWh, charge_state =
 23.0295 kWh, price = \$0.0780
 (d=3.0, t=10.0) | charge = 0.0000 kWh, discharge = 3.7778 kWh, charge_state =
 19.2517 kWh, price = \$0.0898
 (d=3.0, t=11.0) | charge = 0.0000 kWh, discharge = 3.8640 kWh, charge_state =
 15.3877 kWh, price = \$0.0934
 (d=3.0, t=20.0) | charge = 0.0000 kWh, discharge = 2.7255 kWh, charge_state =
 12.6622 kWh, price = \$0.0920
 (d=3.0, t=21.0) | charge = 0.0000 kWh, discharge = 2.8290 kWh, charge_state =
 9.8332 kWh, price = \$0.0869
 (d=3.0, t=22.0) | charge = 0.0000 kWh, discharge = 2.2080 kWh, charge_state =
 7.6252 kWh, price = \$0.0724
 (d=3.0, t=23.0) | charge = 0.0000 kWh, discharge = 2.2252 kWh, charge_state =

5.4000 kWh, price = \$0.0710
 (d=4.0, t=0.0) | charge = 2.1907 kWh, discharge = 2.1907 kWh, charge_state =
 5.4000 kWh, price = \$0.0485
 (d=4.0, t=1.0) | charge = 2.8323 kWh, discharge = 2.1045 kWh, charge_state =
 6.1278 kWh, price = \$0.0408
 (d=4.0, t=2.0) | charge = 5.0000 kWh, discharge = 2.1045 kWh, charge_state =
 9.0233 kWh, price = \$0.0268
 (d=4.0, t=3.0) | charge = 5.0000 kWh, discharge = 1.7423 kWh, charge_state =
 12.2810 kWh, price = \$0.0199
 (d=4.0, t=4.0) | charge = 5.0000 kWh, discharge = 2.0010 kWh, charge_state =
 15.2800 kWh, price = \$0.0174
 (d=4.0, t=5.0) | charge = 5.0000 kWh, discharge = 1.9837 kWh, charge_state =
 18.2963 kWh, price = \$0.0180
 (d=4.0, t=6.0) | charge = 5.0000 kWh, discharge = 2.0010 kWh, charge_state =
 21.2953 kWh, price = \$0.0252
 (d=4.0, t=7.0) | charge = 5.0000 kWh, discharge = 1.9837 kWh, charge_state =
 24.3115 kWh, price = \$0.0271
 (d=4.0, t=8.0) | charge = 5.0000 kWh, discharge = 2.3115 kWh, charge_state =
 27.0000 kWh, price = \$0.0343
 (d=4.0, t=9.0) | charge = 0.0000 kWh, discharge = 2.7945 kWh, charge_state =
 24.2055 kWh, price = \$0.0522
 (d=4.0, t=10.0) | charge = 0.0000 kWh, discharge = 3.5707 kWh, charge_state =
 20.6348 kWh, price = \$0.0694
 (d=4.0, t=11.0) | charge = 0.0000 kWh, discharge = 3.6397 kWh, charge_state =
 16.9950 kWh, price = \$0.0845
 (d=4.0, t=20.0) | charge = 0.0000 kWh, discharge = 2.9670 kWh, charge_state =
 14.0280 kWh, price = \$0.0886
 (d=4.0, t=21.0) | charge = 0.0000 kWh, discharge = 2.5703 kWh, charge_state =
 11.4578 kWh, price = \$0.0849
 (d=4.0, t=22.0) | charge = 0.0000 kWh, discharge = 2.4840 kWh, charge_state =
 8.9738 kWh, price = \$0.0681
 (d=4.0, t=23.0) | charge = 0.0000 kWh, discharge = 2.3460 kWh, charge_state =
 6.6278 kWh, price = \$0.0526
 (d=5.0, t=0.0) | charge = 2.0900 kWh, discharge = 2.3805 kWh, charge_state =
 6.3373 kWh, price = \$0.0414
 (d=5.0, t=1.0) | charge = 5.0000 kWh, discharge = 2.0873 kWh, charge_state =
 9.2500 kWh, price = \$0.0290
 (d=5.0, t=2.0) | charge = 5.0000 kWh, discharge = 2.1045 kWh, charge_state =
 12.1455 kWh, price = \$0.0156
 (d=5.0, t=3.0) | charge = 5.0000 kWh, discharge = 2.2425 kWh, charge_state =
 14.9030 kWh, price = \$0.0076
 (d=5.0, t=4.0) | charge = 5.0000 kWh, discharge = 2.1562 kWh, charge_state =
 17.7468 kWh, price = \$0.0064
 (d=5.0, t=5.0) | charge = 5.0000 kWh, discharge = 2.9670 kWh, charge_state =
 19.7798 kWh, price = \$0.0116
 (d=5.0, t=6.0) | charge = 5.0000 kWh, discharge = 2.5013 kWh, charge_state =
 22.2785 kWh, price = \$0.0240
 (d=5.0, t=7.0) | charge = 5.0000 kWh, discharge = 2.2080 kWh, charge_state =

25.0705 kWh, price = \$0.0257
 (d=5.0, t=8.0) | charge = 5.0000 kWh, discharge = 3.0705 kWh, charge_state =
 27.0000 kWh, price = \$0.0359
 (d=5.0, t=9.0) | charge = 2.8463 kWh, discharge = 2.8462 kWh, charge_state =
 27.0000 kWh, price = \$0.0574
 (d=5.0, t=10.0) | charge = 3.7057 kWh, discharge = 3.7950 kWh, charge_state =
 26.9107 kWh, price = \$0.0691
 (d=5.0, t=11.0) | charge = 0.0000 kWh, discharge = 4.0883 kWh, charge_state =
 22.8225 kWh, price = \$0.0794
 (d=5.0, t=12.0) | charge = 0.0000 kWh, discharge = 3.6915 kWh, charge_state =
 19.1310 kWh, price = \$0.0957
 (d=5.0, t=18.0) | charge = 0.0000 kWh, discharge = 4.0365 kWh, charge_state =
 15.0945 kWh, price = \$0.0990
 (d=5.0, t=19.0) | charge = 0.0000 kWh, discharge = 3.8813 kWh, charge_state =
 11.2132 kWh, price = \$0.0837
 (d=5.0, t=20.0) | charge = 0.0000 kWh, discharge = 3.2258 kWh, charge_state =
 7.9875 kWh, price = \$0.0880
 (d=5.0, t=21.0) | charge = 0.0000 kWh, discharge = 2.5875 kWh, charge_state =
 5.4000 kWh, price = \$0.0818
 (d=5.0, t=22.0) | charge = 2.1735 kWh, discharge = 2.1735 kWh, charge_state =
 5.4000 kWh, price = \$0.0654
 (d=5.0, t=23.0) | charge = 5.0000 kWh, discharge = 2.5013 kWh, charge_state =
 7.8987 kWh, price = \$0.0528
 (d=6.0, t=0.0) | charge = 5.0000 kWh, discharge = 2.0873 kWh, charge_state =
 10.8115 kWh, price = \$0.0444
 (d=6.0, t=1.0) | charge = 5.0000 kWh, discharge = 2.0700 kWh, charge_state =
 13.7415 kWh, price = \$0.0326
 (d=6.0, t=2.0) | charge = 5.0000 kWh, discharge = 2.2770 kWh, charge_state =
 16.4645 kWh, price = \$0.0230
 (d=6.0, t=3.0) | charge = 5.0000 kWh, discharge = 2.1390 kWh, charge_state =
 19.3255 kWh, price = \$0.0194
 (d=6.0, t=4.0) | charge = 5.0000 kWh, discharge = 1.9665 kWh, charge_state =
 22.3590 kWh, price = \$0.0251
 (d=6.0, t=5.0) | charge = 5.0000 kWh, discharge = 3.0015 kWh, charge_state =
 24.3575 kWh, price = \$0.0353
 (d=6.0, t=6.0) | charge = 5.0000 kWh, discharge = 2.3632 kWh, charge_state =
 26.9942 kWh, price = \$0.0538
 (d=6.0, t=7.0) | charge = 2.2828 kWh, discharge = 2.2770 kWh, charge_state =
 27.0000 kWh, price = \$0.0646
 (d=6.0, t=8.0) | charge = 2.6738 kWh, discharge = 2.6738 kWh, charge_state =
 27.0000 kWh, price = \$0.0651
 (d=6.0, t=9.0) | charge = 3.2227 kWh, discharge = 3.5707 kWh, charge_state =
 26.6520 kWh, price = \$0.0710
 (d=6.0, t=10.0) | charge = 0.0000 kWh, discharge = 4.1745 kWh, charge_state =
 22.4775 kWh, price = \$0.0755
 (d=6.0, t=11.0) | charge = 0.0000 kWh, discharge = 4.8990 kWh, charge_state =
 17.5785 kWh, price = \$0.0903
 (d=6.0, t=18.0) | charge = 0.0000 kWh, discharge = 4.4160 kWh, charge_state =

13.1625 kWh, price = \$0.0962
 (d=6.0, t=19.0) | charge = 0.0000 kWh, discharge = 4.3643 kWh, charge_state =
 8.7982 kWh, price = \$0.0812
 (d=6.0, t=20.0) | charge = 0.0000 kWh, discharge = 3.3982 kWh, charge_state =
 5.4000 kWh, price = \$0.0941
 (d=6.0, t=21.0) | charge = 3.0187 kWh, discharge = 3.0187 kWh, charge_state =
 5.4000 kWh, price = \$0.0694
 (d=6.0, t=22.0) | charge = 5.0000 kWh, discharge = 2.6393 kWh, charge_state =
 7.7607 kWh, price = \$0.0630
 (d=6.0, t=23.0) | charge = 5.0000 kWh, discharge = 2.2770 kWh, charge_state =
 10.4837 kWh, price = \$0.0541
 (d=7.0, t=0.0) | charge = 4.2390 kWh, discharge = 2.6565 kWh, charge_state =
 12.0662 kWh, price = \$0.0675
 (d=7.0, t=1.0) | charge = 5.0000 kWh, discharge = 2.1390 kWh, charge_state =
 14.9272 kWh, price = \$0.0611
 (d=7.0, t=2.0) | charge = 5.0000 kWh, discharge = 2.1045 kWh, charge_state =
 17.8227 kWh, price = \$0.0589
 (d=7.0, t=3.0) | charge = 5.0000 kWh, discharge = 2.5185 kWh, charge_state =
 20.3042 kWh, price = \$0.0563
 (d=7.0, t=4.0) | charge = 5.0000 kWh, discharge = 2.0355 kWh, charge_state =
 23.2687 kWh, price = \$0.0601
 (d=7.0, t=5.0) | charge = 5.0000 kWh, discharge = 3.5017 kWh, charge_state =
 24.7670 kWh, price = \$0.0665
 (d=7.0, t=6.0) | charge = 0.0000 kWh, discharge = 2.7428 kWh, charge_state =
 22.0242 kWh, price = \$0.0770
 (d=7.0, t=7.0) | charge = 0.0000 kWh, discharge = 2.4668 kWh, charge_state =
 19.5575 kWh, price = \$0.0858
 (d=7.0, t=8.0) | charge = 0.0000 kWh, discharge = 2.4668 kWh, charge_state =
 17.0907 kWh, price = \$0.0789
 (d=7.0, t=9.0) | charge = 0.0000 kWh, discharge = 2.6910 kWh, charge_state =
 14.3997 kWh, price = \$0.0856
 (d=7.0, t=21.0) | charge = 0.0000 kWh, discharge = 2.8117 kWh, charge_state =
 11.5880 kWh, price = \$0.0917
 (d=7.0, t=22.0) | charge = 0.0000 kWh, discharge = 2.4495 kWh, charge_state =
 9.1385 kWh, price = \$0.0710
 (d=7.0, t=23.0) | charge = 5.0000 kWh, discharge = 2.2080 kWh, charge_state =
 11.9305 kWh, price = \$0.0665
 (d=8.0, t=0.0) | charge = 0.0000 kWh, discharge = 2.1390 kWh, charge_state =
 9.7915 kWh, price = \$0.0716
 (d=8.0, t=1.0) | charge = 5.0000 kWh, discharge = 2.3632 kWh, charge_state =
 12.4282 kWh, price = \$0.0656
 (d=8.0, t=2.0) | charge = 5.0000 kWh, discharge = 2.0355 kWh, charge_state =
 15.3927 kWh, price = \$0.0627
 (d=8.0, t=3.0) | charge = 5.0000 kWh, discharge = 1.9665 kWh, charge_state =
 18.4262 kWh, price = \$0.0606
 (d=8.0, t=4.0) | charge = 5.0000 kWh, discharge = 1.9492 kWh, charge_state =
 21.4770 kWh, price = \$0.0616
 (d=8.0, t=5.0) | charge = 0.0000 kWh, discharge = 2.7255 kWh, charge_state =

18.7515 kWh, price = \$0.0685
 (d=8.0, t=6.0) | charge = 0.0000 kWh, discharge = 2.1562 kWh, charge_state =
 16.5952 kWh, price = \$0.0824
 (d=8.0, t=7.0) | charge = 0.0000 kWh, discharge = 1.8285 kWh, charge_state =
 14.7667 kWh, price = \$0.0871
 (d=8.0, t=8.0) | charge = 0.0000 kWh, discharge = 1.8458 kWh, charge_state =
 12.9210 kWh, price = \$0.0932
 (d=8.0, t=9.0) | charge = 0.0000 kWh, discharge = 1.5525 kWh, charge_state =
 11.3685 kWh, price = \$0.0995
 (d=8.0, t=21.0) | charge = 0.0000 kWh, discharge = 2.2425 kWh, charge_state =
 9.1260 kWh, price = \$0.1017
 (d=8.0, t=22.0) | charge = 0.0000 kWh, discharge = 2.0355 kWh, charge_state =
 7.0905 kWh, price = \$0.0768
 (d=8.0, t=23.0) | charge = 0.0000 kWh, discharge = 1.6905 kWh, charge_state =
 5.4000 kWh, price = \$0.0721
 (d=9.0, t=0.0) | charge = 5.0000 kWh, discharge = 2.0700 kWh, charge_state =
 8.3300 kWh, price = \$0.0575
 (d=9.0, t=1.0) | charge = 5.0000 kWh, discharge = 1.6215 kWh, charge_state =
 11.7085 kWh, price = \$0.0550
 (d=9.0, t=2.0) | charge = 5.0000 kWh, discharge = 1.6732 kWh, charge_state =
 15.0352 kWh, price = \$0.0517
 (d=9.0, t=3.0) | charge = 5.0000 kWh, discharge = 1.6560 kWh, charge_state =
 18.3792 kWh, price = \$0.0500
 (d=9.0, t=4.0) | charge = 5.0000 kWh, discharge = 1.8975 kWh, charge_state =
 21.4817 kWh, price = \$0.0498
 (d=9.0, t=5.0) | charge = 5.0000 kWh, discharge = 1.8113 kWh, charge_state =
 24.6705 kWh, price = \$0.0535
 (d=9.0, t=6.0) | charge = 0.9865 kWh, discharge = 1.8802 kWh, charge_state =
 23.7768 kWh, price = \$0.0619
 (d=9.0, t=7.0) | charge = 5.0000 kWh, discharge = 1.7768 kWh, charge_state =
 27.0000 kWh, price = \$0.0619
 (d=9.0, t=8.0) | charge = 2.1877 kWh, discharge = 2.2252 kWh, charge_state =
 26.9625 kWh, price = \$0.0646
 (d=9.0, t=9.0) | charge = 0.0000 kWh, discharge = 3.1223 kWh, charge_state =
 23.8402 kWh, price = \$0.0726
 (d=9.0, t=10.0) | charge = 0.0000 kWh, discharge = 3.5880 kWh, charge_state =
 20.2522 kWh, price = \$0.0872
 (d=9.0, t=11.0) | charge = 0.0000 kWh, discharge = 2.9152 kWh, charge_state =
 17.3370 kWh, price = \$0.1005
 (d=9.0, t=18.0) | charge = 0.0000 kWh, discharge = 2.9842 kWh, charge_state =
 14.3527 kWh, price = \$0.0890
 (d=9.0, t=19.0) | charge = 0.0000 kWh, discharge = 2.5875 kWh, charge_state =
 11.7652 kWh, price = \$0.0860
 (d=9.0, t=20.0) | charge = 0.0000 kWh, discharge = 2.2080 kWh, charge_state =
 9.5572 kWh, price = \$0.0936
 (d=9.0, t=21.0) | charge = 0.0000 kWh, discharge = 2.4322 kWh, charge_state =
 7.1250 kWh, price = \$0.0837
 (d=9.0, t=22.0) | charge = 0.0000 kWh, discharge = 1.7250 kWh, charge_state =

5.4000 kWh, price = \$0.0689
 (d=9.0, t=23.0) | charge = 1.8457 kWh, discharge = 1.8458 kWh, charge_state =
 5.4000 kWh, price = \$0.0627
 (d=10.0, t=0.0) | charge = 5.0000 kWh, discharge = 1.8802 kWh, charge_state =
 8.5197 kWh, price = \$0.0601
 (d=10.0, t=1.0) | charge = 0.0000 kWh, discharge = 1.5697 kWh, charge_state =
 6.9500 kWh, price = \$0.0624
 (d=10.0, t=2.0) | charge = 5.0000 kWh, discharge = 1.8630 kWh, charge_state =
 10.0870 kWh, price = \$0.0549
 (d=10.0, t=3.0) | charge = 5.0000 kWh, discharge = 1.5870 kWh, charge_state =
 13.5000 kWh, price = \$0.0510
 (d=10.0, t=4.0) | charge = 5.0000 kWh, discharge = 1.5697 kWh, charge_state =
 16.9302 kWh, price = \$0.0500
 (d=10.0, t=5.0) | charge = 5.0000 kWh, discharge = 1.5870 kWh, charge_state =
 20.3432 kWh, price = \$0.0511
 (d=10.0, t=6.0) | charge = 0.0000 kWh, discharge = 1.8802 kWh, charge_state =
 18.4630 kWh, price = \$0.0623
 (d=10.0, t=7.0) | charge = 5.0000 kWh, discharge = 1.5870 kWh, charge_state =
 21.8760 kWh, price = \$0.0593
 (d=10.0, t=8.0) | charge = 5.0000 kWh, discharge = 1.8285 kWh, charge_state =
 25.0475 kWh, price = \$0.0562
 (d=10.0, t=9.0) | charge = 3.2260 kWh, discharge = 2.0183 kWh, charge_state =
 26.2552 kWh, price = \$0.0611
 (d=10.0, t=10.0) | charge = 0.0000 kWh, discharge = 3.0187 kWh, charge_state =
 23.2365 kWh, price = \$0.0746
 (d=10.0, t=11.0) | charge = 0.0000 kWh, discharge = 3.6742 kWh, charge_state =
 19.5622 kWh, price = \$0.0871
 (d=10.0, t=12.0) | charge = 0.0000 kWh, discharge = 3.5707 kWh, charge_state =
 15.9915 kWh, price = \$0.1010
 (d=10.0, t=19.0) | charge = 0.0000 kWh, discharge = 3.0187 kWh, charge_state =
 12.9727 kWh, price = \$0.0849
 (d=10.0, t=20.0) | charge = 0.0000 kWh, discharge = 2.8290 kWh, charge_state =
 10.1437 kWh, price = \$0.0839
 (d=10.0, t=21.0) | charge = 0.0000 kWh, discharge = 2.6048 kWh, charge_state =
 7.5390 kWh, price = \$0.0756
 (d=10.0, t=22.0) | charge = 0.0000 kWh, discharge = 2.1390 kWh, charge_state =
 5.4000 kWh, price = \$0.0632
 (d=10.0, t=23.0) | charge = 2.2598 kWh, discharge = 2.2597 kWh, charge_state =
 5.4000 kWh, price = \$0.0571
 (d=11.0, t=0.0) | charge = 2.1735 kWh, discharge = 2.1735 kWh, charge_state =
 5.4000 kWh, price = \$0.0539
 (d=11.0, t=1.0) | charge = 4.6755 kWh, discharge = 2.0010 kWh, charge_state =
 8.0745 kWh, price = \$0.0454
 (d=11.0, t=2.0) | charge = 5.0000 kWh, discharge = 1.9492 kWh, charge_state =
 11.1253 kWh, price = \$0.0411
 (d=11.0, t=3.0) | charge = 5.0000 kWh, discharge = 1.7250 kWh, charge_state =
 14.4003 kWh, price = \$0.0245
 (d=11.0, t=4.0) | charge = 5.0000 kWh, discharge = 1.8285 kWh, charge_state =

17.5718 kWh, price = \$0.0188
 (d=11.0, t=5.0) | charge = 5.0000 kWh, discharge = 1.9147 kWh, charge_state =
 20.6570 kWh, price = \$0.0254
 (d=11.0, t=6.0) | charge = 5.0000 kWh, discharge = 1.6560 kWh, charge_state =
 24.0010 kWh, price = \$0.0350
 (d=11.0, t=7.0) | charge = 5.0000 kWh, discharge = 2.0010 kWh, charge_state =
 27.0000 kWh, price = \$0.0385
 (d=11.0, t=8.0) | charge = 2.6220 kWh, discharge = 2.6220 kWh, charge_state =
 27.0000 kWh, price = \$0.0463
 (d=11.0, t=9.0) | charge = 3.2430 kWh, discharge = 3.2430 kWh, charge_state =
 27.0000 kWh, price = \$0.0526
 (d=11.0, t=10.0) | charge = 3.4845 kWh, discharge = 3.4845 kWh, charge_state =
 27.0000 kWh, price = \$0.0669
 (d=11.0, t=11.0) | charge = 0.2385 kWh, discharge = 3.4672 kWh, charge_state =
 23.7712 kWh, price = \$0.0755
 (d=11.0, t=12.0) | charge = 0.0000 kWh, discharge = 3.6052 kWh, charge_state =
 20.1660 kWh, price = \$0.0863
 (d=11.0, t=13.0) | charge = 0.0000 kWh, discharge = 4.0020 kWh, charge_state =
 16.1640 kWh, price = \$0.1014
 (d=11.0, t=18.0) | charge = 0.0000 kWh, discharge = 3.7088 kWh, charge_state =
 12.4552 kWh, price = \$0.0923
 (d=11.0, t=19.0) | charge = 0.0000 kWh, discharge = 3.6397 kWh, charge_state =
 8.8155 kWh, price = \$0.0862
 (d=11.0, t=20.0) | charge = 0.0000 kWh, discharge = 3.4155 kWh, charge_state =
 5.4000 kWh, price = \$0.0837
 (d=11.0, t=21.0) | charge = 3.0705 kWh, discharge = 3.0705 kWh, charge_state =
 5.4000 kWh, price = \$0.0692
 (d=11.0, t=22.0) | charge = 2.8290 kWh, discharge = 2.8290 kWh, charge_state =
 5.4000 kWh, price = \$0.0619
 (d=11.0, t=23.0) | charge = 2.5875 kWh, discharge = 2.5875 kWh, charge_state =
 5.4000 kWh, price = \$0.0558
 (d=12.0, t=0.0) | charge = 5.0000 kWh, discharge = 2.5358 kWh, charge_state =
 7.8642 kWh, price = \$0.0472
 (d=12.0, t=1.0) | charge = 5.0000 kWh, discharge = 2.5358 kWh, charge_state =
 10.3285 kWh, price = \$0.0340
 (d=12.0, t=2.0) | charge = 5.0000 kWh, discharge = 2.1735 kWh, charge_state =
 13.1550 kWh, price = \$0.0213
 (d=12.0, t=3.0) | charge = 5.0000 kWh, discharge = 2.1045 kWh, charge_state =
 16.0505 kWh, price = \$0.0144
 (d=12.0, t=4.0) | charge = 5.0000 kWh, discharge = 2.4150 kWh, charge_state =
 18.6355 kWh, price = \$0.0184
 (d=12.0, t=5.0) | charge = 5.0000 kWh, discharge = 3.1395 kWh, charge_state =
 20.4960 kWh, price = \$0.0259
 (d=12.0, t=6.0) | charge = 5.0000 kWh, discharge = 2.6220 kWh, charge_state =
 22.8740 kWh, price = \$0.0450
 (d=12.0, t=7.0) | charge = 5.0000 kWh, discharge = 2.6565 kWh, charge_state =
 25.2175 kWh, price = \$0.0461
 (d=12.0, t=8.0) | charge = 4.1458 kWh, discharge = 2.3632 kWh, charge_state =

27.0000 kWh, price = \$0.0507
 (d=12.0, t=9.0) | charge = 3.2430 kWh, discharge = 3.2430 kWh, charge_state =
 27.0000 kWh, price = \$0.0570
 (d=12.0, t=10.0) | charge = 0.0000 kWh, discharge = 3.9503 kWh, charge_state =
 23.0498 kWh, price = \$0.0744
 (d=12.0, t=11.0) | charge = 0.0000 kWh, discharge = 4.0537 kWh, charge_state =
 18.9960 kWh, price = \$0.0842
 (d=12.0, t=12.0) | charge = 0.0000 kWh, discharge = 4.2263 kWh, charge_state =
 14.7698 kWh, price = \$0.0992
 (d=12.0, t=21.0) | charge = 0.0000 kWh, discharge = 3.0878 kWh, charge_state =
 11.6820 kWh, price = \$0.0802
 (d=12.0, t=22.0) | charge = 0.0000 kWh, discharge = 2.4322 kWh, charge_state =
 9.2498 kWh, price = \$0.0717
 (d=12.0, t=23.0) | charge = 0.0000 kWh, discharge = 2.5013 kWh, charge_state =
 6.7485 kWh, price = \$0.0606
 (d=13.0, t=0.0) | charge = 4.9510 kWh, discharge = 2.4322 kWh, charge_state =
 9.2673 kWh, price = \$0.0577
 (d=13.0, t=1.0) | charge = 5.0000 kWh, discharge = 2.2252 kWh, charge_state =
 12.0420 kWh, price = \$0.0539
 (d=13.0, t=2.0) | charge = 5.0000 kWh, discharge = 2.1218 kWh, charge_state =
 14.9203 kWh, price = \$0.0484
 (d=13.0, t=3.0) | charge = 5.0000 kWh, discharge = 2.4322 kWh, charge_state =
 17.4880 kWh, price = \$0.0428
 (d=13.0, t=4.0) | charge = 5.0000 kWh, discharge = 2.0700 kWh, charge_state =
 20.4180 kWh, price = \$0.0411
 (d=13.0, t=5.0) | charge = 5.0000 kWh, discharge = 3.4500 kWh, charge_state =
 21.9680 kWh, price = \$0.0487
 (d=13.0, t=6.0) | charge = 5.0000 kWh, discharge = 2.3287 kWh, charge_state =
 24.6393 kWh, price = \$0.0577
 (d=13.0, t=7.0) | charge = 5.0000 kWh, discharge = 2.6393 kWh, charge_state =
 27.0000 kWh, price = \$0.0536
 (d=13.0, t=8.0) | charge = 3.1223 kWh, discharge = 3.1223 kWh, charge_state =
 27.0000 kWh, price = \$0.0598
 (d=13.0, t=9.0) | charge = 0.0000 kWh, discharge = 3.8295 kWh, charge_state =
 23.1705 kWh, price = \$0.0661
 (d=13.0, t=10.0) | charge = 0.0000 kWh, discharge = 3.9848 kWh, charge_state =
 19.1858 kWh, price = \$0.0788
 (d=13.0, t=11.0) | charge = 0.0000 kWh, discharge = 4.7265 kWh, charge_state =
 14.4593 kWh, price = \$0.0890
 (d=13.0, t=21.0) | charge = 0.0000 kWh, discharge = 2.5013 kWh, charge_state =
 11.9580 kWh, price = \$0.0853
 (d=13.0, t=22.0) | charge = 0.0000 kWh, discharge = 2.5875 kWh, charge_state =
 9.3705 kWh, price = \$0.0756
 (d=13.0, t=23.0) | charge = 0.0000 kWh, discharge = 2.2252 kWh, charge_state =
 7.1453 kWh, price = \$0.0669
 (d=14.0, t=0.0) | charge = 1.5325 kWh, discharge = 2.3287 kWh, charge_state =
 6.3490 kWh, price = \$0.0650
 (d=14.0, t=1.0) | charge = 5.0000 kWh, discharge = 2.0010 kWh, charge_state =

9.3480 kWh, price = \$0.0586
 (d=14.0, t=2.0) | charge = 5.0000 kWh, discharge = 2.0183 kWh, charge_state =
 12.3297 kWh, price = \$0.0549
 (d=14.0, t=3.0) | charge = 5.0000 kWh, discharge = 2.1218 kWh, charge_state =
 15.2080 kWh, price = \$0.0532
 (d=14.0, t=4.0) | charge = 5.0000 kWh, discharge = 2.0700 kWh, charge_state =
 18.1380 kWh, price = \$0.0529
 (d=14.0, t=5.0) | charge = 5.0000 kWh, discharge = 2.7083 kWh, charge_state =
 20.4297 kWh, price = \$0.0575
 (d=14.0, t=6.0) | charge = 5.0000 kWh, discharge = 2.3977 kWh, charge_state =
 23.0320 kWh, price = \$0.0612
 (d=14.0, t=7.0) | charge = 5.0000 kWh, discharge = 1.9492 kWh, charge_state =
 26.0827 kWh, price = \$0.0639
 (d=14.0, t=8.0) | charge = 0.0000 kWh, discharge = 2.2942 kWh, charge_state =
 23.7885 kWh, price = \$0.0668
 (d=14.0, t=9.0) | charge = 0.0000 kWh, discharge = 3.1395 kWh, charge_state =
 20.6490 kWh, price = \$0.0696
 (d=14.0, t=10.0) | charge = 0.0000 kWh, discharge = 4.0365 kWh, charge_state =
 16.6125 kWh, price = \$0.0801
 (d=14.0, t=11.0) | charge = 0.0000 kWh, discharge = 4.1400 kWh, charge_state =
 12.4725 kWh, price = \$0.0883
 (d=14.0, t=21.0) | charge = 0.0000 kWh, discharge = 2.5013 kWh, charge_state =
 9.9712 kWh, price = \$0.0943
 (d=14.0, t=22.0) | charge = 0.0000 kWh, discharge = 2.2425 kWh, charge_state =
 7.7287 kWh, price = \$0.0818
 (d=14.0, t=23.0) | charge = 0.0000 kWh, discharge = 2.3287 kWh, charge_state =
 5.4000 kWh, price = \$0.0697

```

[7]: import pandas as pd
import gurobipy as gp
from gurobipy import GRB

# Load in Account A data
df = pd.read_csv("account_b.csv")

# Convert Datetime variable so it is parsed correctly
df["Datetime"] = pd.to_datetime(df["Datetime"], format="mixed", errors="coerce")

# Extract date and hour
df["date_only"] = df["Datetime"].dt.date
df["hour"] = df["Datetime"].dt.hour

# First 2 weeks (14 days) only
time = sorted(df["date_only"].unique())[:14]
df_14 = df[df["date_only"].isin(time)].copy()

# Map days and hours indexes

```

```

day_map = {day: i + 1 for i, day in enumerate(time)}
df_14["d"] = df_14["date_only"].map(day_map)
df_14["t"] = df_14["hour"]

# Calculate Price and Quantity
hourly = (
    df_14.groupby(["d", "t"], as_index=False)
    .agg({
        "Quantity": "sum",
        "Settlement.Point.Price.KWH": "mean"
    })
)

Price = {(row["d"], row["t"]): row["Settlement.Point.Price.KWH"] for _, row in hourly.iterrows()}
Quantity = {(row["d"], row["t"]): row["Quantity"] for _, row in hourly.iterrows()}

# Create set indexes
DT = sorted(list(Quantity.keys()))

# Max batteries for 2 week period
B_max = 110

# Battery cost
C_battery = 14900

# Charging/Discharging rate per battery per hour
r_ch = 5
r_dis = 11.5

m = gp.Model("battery_model_account_b")

# Decision Variables
x_B = m.addVar(vtype=GRB.INTEGER, lb=0, ub=B_max)
e = m.addVars(range(1, B_max + 1), vtype=GRB.BINARY)
y = m.addVars(range(1, B_max + 1), DT, vtype=GRB.BINARY)
charge = m.addVars(DT, vtype=GRB.CONTINUOUS, lb=0)
discharge = m.addVars(DT, vtype=GRB.CONTINUOUS, lb=0)
cs = m.addVars(DT, vtype=GRB.CONTINUOUS, lb=0)
charge_b = m.addVars(range(1, B_max+1), DT, lb=0, vtype=GRB.CONTINUOUS)
discharge_b = m.addVars(range(1, B_max+1), DT, lb=0, vtype=GRB.CONTINUOUS)

# Constraints
# sum_b e_{A,b} = x_B
m.addConstr(
    gp.quicksum(e[b] for b in range(1, B_max + 1)) == x_B,

```



```

        name="num_active_batteries"
    )

    #  $y_{b,A,t,d} \leq e_{A,b}$ 
    for b in range(1, B_max + 1):
        for d, t in DT:
            m.addConstr(y[b, d, t] <= e[b])

    #  $g_{\{A,t,d\}} = \text{Quantity}_{\{A,t,d\}}$ 
    for d, t in DT:
        m.addConstr(discharge[d, t] == Quantity[d, t])

    # charging state capacity constraints
    for d, t in DT:
        m.addConstr(cs[d, t] <= 13.5 * x_B)
        m.addConstr(cs[d, t] >= 2.7 * x_B)

    # charging/discharging rate limits
    for d, t in DT:
        m.addConstr(charge[d, t] <= r_ch * x_B)
        m.addConstr(discharge[d, t] <= r_dis * x_B)

    # Charging state across time
    DT_sorted = sorted(DT)

    # Charging state constraints
    for i, (d, t) in enumerate(DT_sorted):
        if i == 0:
            m.addConstr(
                cs[d, t] == 13.5 * x_B + charge[d, t] - discharge[d, t]
            )
        else:
            prev_d, prev_t = DT_sorted[i - 1]
            m.addConstr(
                cs[d, t] == cs[prev_d, prev_t] + charge[d, t] - discharge[d, t]
            )

    # Battery charging constraint
    for b in range(1, B_max+1):
        for d, t in DT:
            m.addConstr(
                charge_b[b, d, t] <= r_ch * y[b, d, t],
                name=f"charge_mode_b{b}_d{d}_t{t}"
            )

    # Battery discharging constraint

```

```

for b in range(1, B_max+1):
    for d, t in DT:
        m.addConstr(
            discharge_b[b, d, t] <= r_dis * (e[b] - y[b, d, t]),
            name=f"discharge_mode_b{b}_d{d}_t{t}"
        )

# Sum of all charge and discharge for each battery must equal the totals
for d, t in DT:
    m.addConstr(
        charge[d, t] == gp.quicksum(charge_b[b, d, t] for b in range(1, B_max+1)),
        name=f"charge_total_d{d}_t{t}"
    )

    m.addConstr(
        discharge[d, t] == gp.quicksum(discharge_b[b, d, t] for b in range(1, B_max+1)),
        name=f"discharge_total_d{d}_t{t}"
    )

# Objective Function
energy_cost = gp.quicksum(
    charge[d, t] * Price[d, t]
    for d, t in DT
)

battery_cost = x_B * C_battery

m.setObjective(energy_cost + battery_cost, GRB.MINIMIZE)

# Optimize
m.optimize()

# Print results
if m.status == GRB.OPTIMAL:
    print("\nOptimal Solution")
    print(f"x_B = {x_B.X}")
    print(f"Energy cost = {energy_cost.getValue():.4f}")
    print(f"Battery cost = {battery_cost.getValue():.4f}")
    print(f"Total Cost = {m.objVal:.4f}")

    # Display charge/discharge values for some iterations just to check
    print("\nSample hourly values:")
    for d, t in DT_sorted[:600]:
        print(

```

```

        f"(d={d}, t={t}) | charge = {charge[d,t].X:.4f} kWh, "
        f"discharge = {discharge[d,t].X:.4f} kWh, charge_state = {cs[d,t].X:
↪.4f} kWh, price = ${Price[(d,t)]:.4f}"
    )
else:
    print("No optimal solution found.")

```

Gurobi Optimizer version 12.0.0 build v12.0.0rc1 (win64 - Windows 11+.0 (26200.2))

CPU model: Intel(R) Core(TM) Ultra 7 155U, instruction set [SSE2|AVX|AVX2]
Thread count: 12 physical cores, 14 logical processors, using up to 14 threads

Optimize a model with 78079 rows, 77034 columns and 232266 nonzeros

Model fingerprint: 0x1651188f

Variable types: 51513 continuous, 25521 integer (25520 binary)

Coefficient statistics:

```

Matrix range      [1e+00, 1e+01]
Objective range   [6e-03, 1e+04]
Bounds range      [1e+00, 1e+02]
RHS range         [9e+01, 3e+02]

```

Presolve removed 9046 rows and 462 columns

Presolve time: 0.92s

Presolved: 69033 rows, 76572 columns, 214173 nonzeros

Variable types: 51051 continuous, 25521 integer (25520 binary)

Found heuristic solution: objective 1640821.6724

Found heuristic solution: objective 1640820.9606

Deterministic concurrent LP optimizer: primal and dual simplex

Showing primal log only...

Concurrent spin time: 0.01s

Solved with dual simplex

Use crossover to convert LP symmetric solution to basic solution...

Root simplex log...

Iteration	Objective	Primal Inf.	Dual Inf.	Time
30592	7.7947431e+05	0.000000e+00	1.011742e+02	5s
31175	7.7947434e+05	0.000000e+00	0.000000e+00	8s

Extra simplex iterations after uncrush: 3831

Root relaxation: objective 7.794743e+05, 31175 iterations, 6.15 seconds (3.02 work units)

Total elapsed time = 39.55s (DegenMoves)

Total elapsed time = 67.59s (DegenMoves)

Total elapsed time = 74.69s (DegenMoves)

Total elapsed time = 78.13s (DegenMoves)

Nodes			Current Node			Objective Bounds			Work	
Expl	Unexpl		Obj	Depth	IntInf	Incumbent	BestBd	Gap	It/Node	Time
	0	0	779474.339	0	169	1640820.96	779474.339	52.5%	-	78s
H	0	0				791265.03956	779474.339	1.49%	-	79s
H	0	0				791265.00506	779474.339	1.49%	-	79s
H	0	0				791262.18114	779474.339	1.49%	-	82s

Cutting planes:

Gomory: 7
MIR: 198
StrongCG: 1
Flow cover: 119
RLT: 1
Relax-and-lift: 6

Explored 1 nodes (104430 simplex iterations) in 83.00 seconds (39.16 work units)
Thread count was 14 (of 14 available processors)

Solution count 5: 791262 791265 791265 ... 1.64082e+06

Optimal solution found (tolerance 1.00e-04)

Best objective 7.912621811417e+05, best bound 7.912588598471e+05, gap 0.0004%

Optimal Solution

x_B = 53.0
Energy cost = 1,562.1811
Battery cost = 789,700.0000
Total Cost = 791,262.1811

Sample hourly values:

(d=1.0, t=0.0) | charge = 0.0000 kWh, discharge = 97.3590 kWh, charge_state = 618.1410 kWh, price = \$0.0613
(d=1.0, t=1.0) | charge = 0.0000 kWh, discharge = 94.0470 kWh, charge_state = 524.0940 kWh, price = \$0.0577
(d=1.0, t=2.0) | charge = 46.4500 kWh, discharge = 96.8760 kWh, charge_state = 473.6680 kWh, price = \$0.0536
(d=1.0, t=3.0) | charge = 220.0000 kWh, discharge = 98.8080 kWh, charge_state = 594.8600 kWh, price = \$0.0500
(d=1.0, t=4.0) | charge = 220.0000 kWh, discharge = 99.3600 kWh, charge_state = 715.5000 kWh, price = \$0.0489
(d=1.0, t=5.0) | charge = 105.6390 kWh, discharge = 105.6390 kWh, charge_state = 715.5000 kWh, price = \$0.0542
(d=1.0, t=6.0) | charge = 143.0370 kWh, discharge = 143.0370 kWh, charge_state = 715.5000 kWh, price = \$0.0607
(d=1.0, t=7.0) | charge = 130.2870 kWh, discharge = 161.7360 kWh, charge_state =

684.0510 kWh, price = \$0.0622
 (d=1.0, t=8.0) | charge = 190.0000 kWh, discharge = 167.1180 kWh, charge_state = 706.9330 kWh, price = \$0.0574
 (d=1.0, t=9.0) | charge = 185.0000 kWh, discharge = 176.4330 kWh, charge_state = 715.5000 kWh, price = \$0.0613
 (d=1.0, t=10.0) | charge = 180.0000 kWh, discharge = 185.5410 kWh, charge_state = 709.9590 kWh, price = \$0.0630
 (d=1.0, t=11.0) | charge = 180.0000 kWh, discharge = 194.5800 kWh, charge_state = 695.3790 kWh, price = \$0.0711
 (d=1.0, t=12.0) | charge = 161.7330 kWh, discharge = 183.1260 kWh, charge_state = 673.9860 kWh, price = \$0.0739
 (d=1.0, t=13.0) | charge = 0.0000 kWh, discharge = 178.6410 kWh, charge_state = 495.3450 kWh, price = \$0.0826
 (d=1.0, t=14.0) | charge = 0.0000 kWh, discharge = 169.3950 kWh, charge_state = 325.9500 kWh, price = \$0.0999
 (d=1.0, t=21.0) | charge = 0.0000 kWh, discharge = 91.8390 kWh, charge_state = 234.1110 kWh, price = \$0.0936
 (d=1.0, t=22.0) | charge = 0.0000 kWh, discharge = 91.0110 kWh, charge_state = 143.1000 kWh, price = \$0.0781
 (d=1.0, t=23.0) | charge = 87.4230 kWh, discharge = 87.4230 kWh, charge_state = 143.1000 kWh, price = \$0.0647
 (d=2.0, t=0.0) | charge = 87.6300 kWh, discharge = 87.6300 kWh, charge_state = 143.1000 kWh, price = \$0.0640
 (d=2.0, t=1.0) | charge = 116.9670 kWh, discharge = 87.9060 kWh, charge_state = 172.1610 kWh, price = \$0.0594
 (d=2.0, t=2.0) | charge = 225.0000 kWh, discharge = 86.7330 kWh, charge_state = 310.4280 kWh, price = \$0.0499
 (d=2.0, t=3.0) | charge = 225.0000 kWh, discharge = 86.9400 kWh, charge_state = 448.4880 kWh, price = \$0.0443
 (d=2.0, t=4.0) | charge = 225.0000 kWh, discharge = 91.1490 kWh, charge_state = 582.3390 kWh, price = \$0.0502
 (d=2.0, t=5.0) | charge = 225.0000 kWh, discharge = 91.8390 kWh, charge_state = 715.5000 kWh, price = \$0.0565
 (d=2.0, t=6.0) | charge = 14.4350 kWh, discharge = 128.3400 kWh, charge_state = 601.5950 kWh, price = \$0.0720
 (d=2.0, t=7.0) | charge = 200.0000 kWh, discharge = 138.5520 kWh, charge_state = 663.0430 kWh, price = \$0.0712
 (d=2.0, t=8.0) | charge = 195.0000 kWh, discharge = 150.9030 kWh, charge_state = 707.1400 kWh, price = \$0.0628
 (d=2.0, t=9.0) | charge = 185.0000 kWh, discharge = 176.6400 kWh, charge_state = 715.5000 kWh, price = \$0.0652
 (d=2.0, t=10.0) | charge = 0.0000 kWh, discharge = 184.1610 kWh, charge_state = 531.3390 kWh, price = \$0.0771
 (d=2.0, t=11.0) | charge = 0.0000 kWh, discharge = 183.1260 kWh, charge_state = 348.2130 kWh, price = \$0.0947
 (d=2.0, t=20.0) | charge = 0.0000 kWh, discharge = 93.7020 kWh, charge_state = 254.5110 kWh, price = \$0.1006
 (d=2.0, t=21.0) | charge = 0.0000 kWh, discharge = 89.0100 kWh, charge_state =

165.5010 kWh, price = \$0.0835
 (d=2.0, t=22.0) | charge = 64.4700 kWh, discharge = 86.8710 kWh, charge_state =
 143.1000 kWh, price = \$0.0765
 (d=2.0, t=23.0) | charge = 88.3200 kWh, discharge = 88.3200 kWh, charge_state =
 143.1000 kWh, price = \$0.0706
 (d=3.0, t=0.0) | charge = 87.3540 kWh, discharge = 87.3540 kWh, charge_state =
 143.1000 kWh, price = \$0.0603
 (d=3.0, t=1.0) | charge = 87.6990 kWh, discharge = 87.6990 kWh, charge_state =
 143.1000 kWh, price = \$0.0507
 (d=3.0, t=2.0) | charge = 225.0000 kWh, discharge = 86.4570 kWh, charge_state =
 281.6430 kWh, price = \$0.0366
 (d=3.0, t=3.0) | charge = 225.0000 kWh, discharge = 86.5260 kWh, charge_state =
 420.1170 kWh, price = \$0.0301
 (d=3.0, t=4.0) | charge = 225.0000 kWh, discharge = 90.0450 kWh, charge_state =
 555.0720 kWh, price = \$0.0274
 (d=3.0, t=5.0) | charge = 225.0000 kWh, discharge = 88.2510 kWh, charge_state =
 691.8210 kWh, price = \$0.0319
 (d=3.0, t=6.0) | charge = 112.5510 kWh, discharge = 88.8720 kWh, charge_state =
 715.5000 kWh, price = \$0.0424
 (d=3.0, t=7.0) | charge = 135.9300 kWh, discharge = 135.9300 kWh, charge_state =
 715.5000 kWh, price = \$0.0449
 (d=3.0, t=8.0) | charge = 135.0000 kWh, discharge = 136.3440 kWh, charge_state =
 714.1560 kWh, price = \$0.0508
 (d=3.0, t=9.0) | charge = 58.0170 kWh, discharge = 143.6580 kWh, charge_state =
 628.5150 kWh, price = \$0.0780
 (d=3.0, t=10.0) | charge = 0.0000 kWh, discharge = 148.7640 kWh, charge_state =
 479.7510 kWh, price = \$0.0898
 (d=3.0, t=11.0) | charge = 0.0000 kWh, discharge = 147.1770 kWh, charge_state =
 332.5740 kWh, price = \$0.0934
 (d=3.0, t=20.0) | charge = 0.0000 kWh, discharge = 96.5310 kWh, charge_state =
 236.0430 kWh, price = \$0.0920
 (d=3.0, t=21.0) | charge = 0.0000 kWh, discharge = 92.9430 kWh, charge_state =
 143.1000 kWh, price = \$0.0869
 (d=3.0, t=22.0) | charge = 91.3560 kWh, discharge = 91.3560 kWh, charge_state =
 143.1000 kWh, price = \$0.0724
 (d=3.0, t=23.0) | charge = 89.7000 kWh, discharge = 89.7000 kWh, charge_state =
 143.1000 kWh, price = \$0.0710
 (d=4.0, t=0.0) | charge = 91.5630 kWh, discharge = 91.5630 kWh, charge_state =
 143.1000 kWh, price = \$0.0485
 (d=4.0, t=1.0) | charge = 89.9760 kWh, discharge = 89.9760 kWh, charge_state =
 143.1000 kWh, price = \$0.0408
 (d=4.0, t=2.0) | charge = 220.0000 kWh, discharge = 93.6330 kWh, charge_state =
 269.4670 kWh, price = \$0.0268
 (d=4.0, t=3.0) | charge = 220.0000 kWh, discharge = 92.2530 kWh, charge_state =
 397.2140 kWh, price = \$0.0199
 (d=4.0, t=4.0) | charge = 225.0000 kWh, discharge = 91.3560 kWh, charge_state =
 530.8580 kWh, price = \$0.0174
 (d=4.0, t=5.0) | charge = 220.0000 kWh, discharge = 101.0160 kWh, charge_state =

649.8420 kWh, price = \$0.0180
 (d=4.0, t=6.0) | charge = 195.0000 kWh, discharge = 157.5270 kWh, charge_state =
 687.3150 kWh, price = \$0.0252
 (d=4.0, t=7.0) | charge = 169.4280 kWh, discharge = 141.2430 kWh, charge_state =
 715.5000 kWh, price = \$0.0271
 (d=4.0, t=8.0) | charge = 137.5170 kWh, discharge = 137.5170 kWh, charge_state =
 715.5000 kWh, price = \$0.0343
 (d=4.0, t=9.0) | charge = 150.3510 kWh, discharge = 150.3510 kWh, charge_state =
 715.5000 kWh, price = \$0.0522
 (d=4.0, t=10.0) | charge = 0.0000 kWh, discharge = 154.6290 kWh, charge_state =
 560.8710 kWh, price = \$0.0694
 (d=4.0, t=11.0) | charge = 0.0000 kWh, discharge = 153.3870 kWh, charge_state =
 407.4840 kWh, price = \$0.0845
 (d=4.0, t=20.0) | charge = 0.0000 kWh, discharge = 108.2610 kWh, charge_state =
 299.2230 kWh, price = \$0.0886
 (d=4.0, t=21.0) | charge = 0.0000 kWh, discharge = 108.5370 kWh, charge_state =
 190.6860 kWh, price = \$0.0849
 (d=4.0, t=22.0) | charge = 61.1580 kWh, discharge = 108.7440 kWh, charge_state =
 143.1000 kWh, price = \$0.0681
 (d=4.0, t=23.0) | charge = 107.4330 kWh, discharge = 107.4330 kWh, charge_state
 = 143.1000 kWh, price = \$0.0526
 (d=5.0, t=0.0) | charge = 107.1570 kWh, discharge = 107.1570 kWh, charge_state =
 143.1000 kWh, price = \$0.0414
 (d=5.0, t=1.0) | charge = 160.0000 kWh, discharge = 110.5380 kWh, charge_state =
 192.5620 kWh, price = \$0.0290
 (d=5.0, t=2.0) | charge = 215.0000 kWh, discharge = 107.2260 kWh, charge_state =
 300.3360 kWh, price = \$0.0156
 (d=5.0, t=3.0) | charge = 215.0000 kWh, discharge = 107.2260 kWh, charge_state =
 408.1100 kWh, price = \$0.0076
 (d=5.0, t=4.0) | charge = 215.0000 kWh, discharge = 107.5710 kWh, charge_state =
 515.5390 kWh, price = \$0.0064
 (d=5.0, t=5.0) | charge = 215.0000 kWh, discharge = 110.2620 kWh, charge_state =
 620.2770 kWh, price = \$0.0116
 (d=5.0, t=6.0) | charge = 195.0000 kWh, discharge = 157.6650 kWh, charge_state =
 657.6120 kWh, price = \$0.0240
 (d=5.0, t=7.0) | charge = 200.0000 kWh, discharge = 144.6240 kWh, charge_state =
 712.9880 kWh, price = \$0.0257
 (d=5.0, t=8.0) | charge = 141.9610 kWh, discharge = 139.4490 kWh, charge_state =
 715.5000 kWh, price = \$0.0359
 (d=5.0, t=9.0) | charge = 142.6920 kWh, discharge = 142.6920 kWh, charge_state =
 715.5000 kWh, price = \$0.0574
 (d=5.0, t=10.0) | charge = 152.2830 kWh, discharge = 152.2830 kWh, charge_state
 = 715.5000 kWh, price = \$0.0691
 (d=5.0, t=11.0) | charge = 155.1810 kWh, discharge = 155.1810 kWh, charge_state
 = 715.5000 kWh, price = \$0.0794
 (d=5.0, t=12.0) | charge = 0.0000 kWh, discharge = 153.4560 kWh, charge_state =
 562.0440 kWh, price = \$0.0957
 (d=5.0, t=18.0) | charge = 0.0000 kWh, discharge = 137.5170 kWh, charge_state =

424.5270 kWh, price = \$0.0990
 (d=5.0, t=19.0) | charge = 0.0000 kWh, discharge = 111.3660 kWh, charge_state = 313.1610 kWh, price = \$0.0837
 (d=5.0, t=20.0) | charge = 0.0000 kWh, discharge = 111.2280 kWh, charge_state = 201.9330 kWh, price = \$0.0880
 (d=5.0, t=21.0) | charge = 51.3600 kWh, discharge = 110.1930 kWh, charge_state = 143.1000 kWh, price = \$0.0818
 (d=5.0, t=22.0) | charge = 108.3300 kWh, discharge = 108.3300 kWh, charge_state = 143.1000 kWh, price = \$0.0654
 (d=5.0, t=23.0) | charge = 109.7100 kWh, discharge = 109.7100 kWh, charge_state = 143.1000 kWh, price = \$0.0528
 (d=6.0, t=0.0) | charge = 150.6230 kWh, discharge = 110.2620 kWh, charge_state = 183.4610 kWh, price = \$0.0444
 (d=6.0, t=1.0) | charge = 215.0000 kWh, discharge = 108.2610 kWh, charge_state = 290.2000 kWh, price = \$0.0326
 (d=6.0, t=2.0) | charge = 215.0000 kWh, discharge = 108.3300 kWh, charge_state = 396.8700 kWh, price = \$0.0230
 (d=6.0, t=3.0) | charge = 215.0000 kWh, discharge = 108.6060 kWh, charge_state = 503.2640 kWh, price = \$0.0194
 (d=6.0, t=4.0) | charge = 215.0000 kWh, discharge = 108.8130 kWh, charge_state = 609.4510 kWh, price = \$0.0251
 (d=6.0, t=5.0) | charge = 215.0000 kWh, discharge = 108.9510 kWh, charge_state = 715.5000 kWh, price = \$0.0353
 (d=6.0, t=6.0) | charge = 180.0000 kWh, discharge = 194.8560 kWh, charge_state = 700.6440 kWh, price = \$0.0538
 (d=6.0, t=7.0) | charge = 175.0000 kWh, discharge = 205.3440 kWh, charge_state = 670.3000 kWh, price = \$0.0646
 (d=6.0, t=8.0) | charge = 175.0000 kWh, discharge = 198.6510 kWh, charge_state = 646.6490 kWh, price = \$0.0651
 (d=6.0, t=9.0) | charge = 170.0000 kWh, discharge = 208.3110 kWh, charge_state = 608.3380 kWh, price = \$0.0710
 (d=6.0, t=10.0) | charge = 160.0000 kWh, discharge = 230.8050 kWh, charge_state = 537.5330 kWh, price = \$0.0755
 (d=6.0, t=11.0) | charge = 0.0000 kWh, discharge = 228.4590 kWh, charge_state = 309.0740 kWh, price = \$0.0903
 (d=6.0, t=18.0) | charge = 0.0000 kWh, discharge = 158.7000 kWh, charge_state = 150.3740 kWh, price = \$0.0962
 (d=6.0, t=19.0) | charge = 211.6440 kWh, discharge = 111.2280 kWh, charge_state = 250.7900 kWh, price = \$0.0812
 (d=6.0, t=20.0) | charge = 0.0000 kWh, discharge = 107.0880 kWh, charge_state = 143.7020 kWh, price = \$0.0941
 (d=6.0, t=21.0) | charge = 100.0000 kWh, discharge = 100.6020 kWh, charge_state = 143.1000 kWh, price = \$0.0694
 (d=6.0, t=22.0) | charge = 150.5320 kWh, discharge = 98.0490 kWh, charge_state = 195.5830 kWh, price = \$0.0630
 (d=6.0, t=23.0) | charge = 220.0000 kWh, discharge = 97.4280 kWh, charge_state = 318.1550 kWh, price = \$0.0541
 (d=7.0, t=0.0) | charge = 0.0000 kWh, discharge = 97.7730 kWh, charge_state =

220.3820 kWh, price = \$0.0675
 (d=7.0, t=1.0) | charge = 220.0000 kWh, discharge = 95.7030 kWh, charge_state =
 344.6790 kWh, price = \$0.0611
 (d=7.0, t=2.0) | charge = 220.0000 kWh, discharge = 96.5310 kWh, charge_state =
 468.1480 kWh, price = \$0.0589
 (d=7.0, t=3.0) | charge = 220.0000 kWh, discharge = 97.7040 kWh, charge_state =
 590.4440 kWh, price = \$0.0563
 (d=7.0, t=4.0) | charge = 220.0000 kWh, discharge = 94.9440 kWh, charge_state =
 715.5000 kWh, price = \$0.0601
 (d=7.0, t=5.0) | charge = 119.5770 kWh, discharge = 119.5770 kWh, charge_state =
 715.5000 kWh, price = \$0.0665
 (d=7.0, t=6.0) | charge = 177.0540 kWh, discharge = 177.0540 kWh, charge_state =
 715.5000 kWh, price = \$0.0770
 (d=7.0, t=7.0) | charge = 0.0000 kWh, discharge = 191.9580 kWh, charge_state =
 523.5420 kWh, price = \$0.0858
 (d=7.0, t=8.0) | charge = 123.0510 kWh, discharge = 189.1980 kWh, charge_state =
 457.3950 kWh, price = \$0.0789
 (d=7.0, t=9.0) | charge = 0.0000 kWh, discharge = 214.8660 kWh, charge_state =
 242.5290 kWh, price = \$0.0856
 (d=7.0, t=21.0) | charge = 0.0000 kWh, discharge = 99.4290 kWh, charge_state =
 143.1000 kWh, price = \$0.0917
 (d=7.0, t=22.0) | charge = 99.0150 kWh, discharge = 99.0150 kWh, charge_state =
 143.1000 kWh, price = \$0.0710
 (d=7.0, t=23.0) | charge = 220.0000 kWh, discharge = 96.9450 kWh, charge_state =
 266.1550 kWh, price = \$0.0665
 (d=8.0, t=0.0) | charge = 0.0000 kWh, discharge = 96.9450 kWh, charge_state =
 169.2100 kWh, price = \$0.0716
 (d=8.0, t=1.0) | charge = 220.0000 kWh, discharge = 94.0470 kWh, charge_state =
 295.1630 kWh, price = \$0.0656
 (d=8.0, t=2.0) | charge = 225.0000 kWh, discharge = 90.3900 kWh, charge_state =
 429.7730 kWh, price = \$0.0627
 (d=8.0, t=3.0) | charge = 225.0000 kWh, discharge = 89.0100 kWh, charge_state =
 565.7630 kWh, price = \$0.0606
 (d=8.0, t=4.0) | charge = 225.0000 kWh, discharge = 90.0450 kWh, charge_state =
 700.7180 kWh, price = \$0.0616
 (d=8.0, t=5.0) | charge = 121.1800 kWh, discharge = 106.3980 kWh, charge_state =
 715.5000 kWh, price = \$0.0685
 (d=8.0, t=6.0) | charge = 151.4550 kWh, discharge = 151.4550 kWh, charge_state =
 715.5000 kWh, price = \$0.0824
 (d=8.0, t=7.0) | charge = 22.1860 kWh, discharge = 147.3840 kWh, charge_state =
 590.3020 kWh, price = \$0.0871
 (d=8.0, t=8.0) | charge = 0.0000 kWh, discharge = 179.4000 kWh, charge_state =
 410.9020 kWh, price = \$0.0932
 (d=8.0, t=9.0) | charge = 0.0000 kWh, discharge = 168.6360 kWh, charge_state =
 242.2660 kWh, price = \$0.0995
 (d=8.0, t=21.0) | charge = 0.0000 kWh, discharge = 98.5320 kWh, charge_state =
 143.7340 kWh, price = \$0.1017
 (d=8.0, t=22.0) | charge = 95.0000 kWh, discharge = 95.6340 kWh, charge_state =

143.1000 kWh, price = \$0.0768
 (d=8.0, t=23.0) | charge = 93.9090 kWh, discharge = 93.9090 kWh, charge_state =
 143.1000 kWh, price = \$0.0721
 (d=9.0, t=0.0) | charge = 95.1510 kWh, discharge = 95.1510 kWh, charge_state =
 143.1000 kWh, price = \$0.0575
 (d=9.0, t=1.0) | charge = 142.3910 kWh, discharge = 93.0120 kWh, charge_state =
 192.4790 kWh, price = \$0.0550
 (d=9.0, t=2.0) | charge = 225.0000 kWh, discharge = 91.6320 kWh, charge_state =
 325.8470 kWh, price = \$0.0517
 (d=9.0, t=3.0) | charge = 225.0000 kWh, discharge = 90.3900 kWh, charge_state =
 460.4570 kWh, price = \$0.0500
 (d=9.0, t=4.0) | charge = 225.0000 kWh, discharge = 87.6300 kWh, charge_state =
 597.8270 kWh, price = \$0.0498
 (d=9.0, t=5.0) | charge = 220.0000 kWh, discharge = 102.3270 kWh, charge_state =
 715.5000 kWh, price = \$0.0535
 (d=9.0, t=6.0) | charge = 118.2230 kWh, discharge = 159.5970 kWh, charge_state =
 674.1260 kWh, price = \$0.0619
 (d=9.0, t=7.0) | charge = 190.0000 kWh, discharge = 148.6260 kWh, charge_state =
 715.5000 kWh, price = \$0.0619
 (d=9.0, t=8.0) | charge = 162.9090 kWh, discharge = 162.9090 kWh, charge_state =
 715.5000 kWh, price = \$0.0646
 (d=9.0, t=9.0) | charge = 171.3960 kWh, discharge = 171.3960 kWh, charge_state =
 715.5000 kWh, price = \$0.0726
 (d=9.0, t=10.0) | charge = 0.0000 kWh, discharge = 183.0570 kWh, charge_state =
 532.4430 kWh, price = \$0.0872
 (d=9.0, t=11.0) | charge = 0.0000 kWh, discharge = 190.0260 kWh, charge_state =
 342.4170 kWh, price = \$0.1005
 (d=9.0, t=18.0) | charge = 0.0000 kWh, discharge = 114.5400 kWh, charge_state =
 227.8770 kWh, price = \$0.0890
 (d=9.0, t=19.0) | charge = 96.2100 kWh, discharge = 90.3900 kWh, charge_state =
 233.6970 kWh, price = \$0.0860
 (d=9.0, t=20.0) | charge = 0.0000 kWh, discharge = 90.5970 kWh, charge_state =
 143.1000 kWh, price = \$0.0936
 (d=9.0, t=21.0) | charge = 91.7700 kWh, discharge = 91.7700 kWh, charge_state =
 143.1000 kWh, price = \$0.0837
 (d=9.0, t=22.0) | charge = 90.5970 kWh, discharge = 90.5970 kWh, charge_state =
 143.1000 kWh, price = \$0.0689
 (d=9.0, t=23.0) | charge = 90.5280 kWh, discharge = 90.5280 kWh, charge_state =
 143.1000 kWh, price = \$0.0627
 (d=10.0, t=0.0) | charge = 186.6450 kWh, discharge = 94.6680 kWh, charge_state =
 235.0770 kWh, price = \$0.0601
 (d=10.0, t=1.0) | charge = 0.0000 kWh, discharge = 91.9770 kWh, charge_state =
 143.1000 kWh, price = \$0.0624
 (d=10.0, t=2.0) | charge = 225.0000 kWh, discharge = 90.3900 kWh, charge_state =
 277.7100 kWh, price = \$0.0549
 (d=10.0, t=3.0) | charge = 225.0000 kWh, discharge = 86.8710 kWh, charge_state =
 415.8390 kWh, price = \$0.0510
 (d=10.0, t=4.0) | charge = 225.0000 kWh, discharge = 87.2850 kWh, charge_state =

553.5540 kWh, price = \$0.0500
 (d=10.0, t=5.0) | charge = 225.0000 kWh, discharge = 87.1470 kWh, charge_state = 691.4070 kWh, price = \$0.0511
 (d=10.0, t=6.0) | charge = 0.0000 kWh, discharge = 86.1810 kWh, charge_state = 605.2260 kWh, price = \$0.0623
 (d=10.0, t=7.0) | charge = 166.8530 kWh, discharge = 130.4790 kWh, charge_state = 641.6000 kWh, price = \$0.0593
 (d=10.0, t=8.0) | charge = 205.0000 kWh, discharge = 131.1000 kWh, charge_state = 715.5000 kWh, price = \$0.0562
 (d=10.0, t=9.0) | charge = 136.6200 kWh, discharge = 136.6200 kWh, charge_state = 715.5000 kWh, price = \$0.0611
 (d=10.0, t=10.0) | charge = 140.0000 kWh, discharge = 142.4850 kWh, charge_state = 713.0150 kWh, price = \$0.0746
 (d=10.0, t=11.0) | charge = 0.0000 kWh, discharge = 151.8690 kWh, charge_state = 561.1460 kWh, price = \$0.0871
 (d=10.0, t=12.0) | charge = 0.0000 kWh, discharge = 117.7830 kWh, charge_state = 443.3630 kWh, price = \$0.1010
 (d=10.0, t=19.0) | charge = 0.0000 kWh, discharge = 133.7220 kWh, charge_state = 309.6410 kWh, price = \$0.0849
 (d=10.0, t=20.0) | charge = 0.0000 kWh, discharge = 135.1020 kWh, charge_state = 174.5390 kWh, price = \$0.0839
 (d=10.0, t=21.0) | charge = 101.5240 kWh, discharge = 132.9630 kWh, charge_state = 143.1000 kWh, price = \$0.0756
 (d=10.0, t=22.0) | charge = 161.3530 kWh, discharge = 132.7560 kWh, charge_state = 171.6970 kWh, price = \$0.0632
 (d=10.0, t=23.0) | charge = 205.0000 kWh, discharge = 132.8940 kWh, charge_state = 243.8030 kWh, price = \$0.0571
 (d=11.0, t=0.0) | charge = 205.0000 kWh, discharge = 133.3080 kWh, charge_state = 315.4950 kWh, price = \$0.0539
 (d=11.0, t=1.0) | charge = 205.0000 kWh, discharge = 132.7560 kWh, charge_state = 387.7390 kWh, price = \$0.0454
 (d=11.0, t=2.0) | charge = 205.0000 kWh, discharge = 133.1010 kWh, charge_state = 459.6380 kWh, price = \$0.0411
 (d=11.0, t=3.0) | charge = 205.0000 kWh, discharge = 136.5510 kWh, charge_state = 528.0870 kWh, price = \$0.0245
 (d=11.0, t=4.0) | charge = 205.0000 kWh, discharge = 133.5150 kWh, charge_state = 599.5720 kWh, price = \$0.0188
 (d=11.0, t=5.0) | charge = 205.0000 kWh, discharge = 133.3080 kWh, charge_state = 671.2640 kWh, price = \$0.0254
 (d=11.0, t=6.0) | charge = 180.0000 kWh, discharge = 184.9200 kWh, charge_state = 666.3440 kWh, price = \$0.0350
 (d=11.0, t=7.0) | charge = 190.0000 kWh, discharge = 169.6710 kWh, charge_state = 686.6730 kWh, price = \$0.0385
 (d=11.0, t=8.0) | charge = 190.0000 kWh, discharge = 171.2580 kWh, charge_state = 705.4150 kWh, price = \$0.0463
 (d=11.0, t=9.0) | charge = 185.0000 kWh, discharge = 174.9150 kWh, charge_state = 715.5000 kWh, price = \$0.0526
 (d=11.0, t=10.0) | charge = 178.7100 kWh, discharge = 178.7100 kWh, charge_state

= 715.5000 kWh, price = \$0.0669
 (d=11.0, t=11.0) | charge = 180.0000 kWh, discharge = 180.1590 kWh, charge_state
 = 715.3410 kWh, price = \$0.0755
 (d=11.0, t=12.0) | charge = 0.0000 kWh, discharge = 183.8160 kWh, charge_state =
 531.5250 kWh, price = \$0.0863
 (d=11.0, t=13.0) | charge = 0.0000 kWh, discharge = 185.1960 kWh, charge_state =
 346.3290 kWh, price = \$0.1014
 (d=11.0, t=18.0) | charge = 0.0000 kWh, discharge = 158.4240 kWh, charge_state =
 187.9050 kWh, price = \$0.0923
 (d=11.0, t=19.0) | charge = 88.7790 kWh, discharge = 133.5840 kWh, charge_state
 = 143.1000 kWh, price = \$0.0862
 (d=11.0, t=20.0) | charge = 134.4120 kWh, discharge = 134.4120 kWh, charge_state
 = 143.1000 kWh, price = \$0.0837
 (d=11.0, t=21.0) | charge = 137.6230 kWh, discharge = 135.6540 kWh, charge_state
 = 145.0690 kWh, price = \$0.0692
 (d=11.0, t=22.0) | charge = 205.0000 kWh, discharge = 131.8590 kWh, charge_state
 = 218.2100 kWh, price = \$0.0619
 (d=11.0, t=23.0) | charge = 205.0000 kWh, discharge = 133.4460 kWh, charge_state
 = 289.7640 kWh, price = \$0.0558
 (d=12.0, t=0.0) | charge = 205.0000 kWh, discharge = 132.5490 kWh, charge_state
 = 362.2150 kWh, price = \$0.0472
 (d=12.0, t=1.0) | charge = 205.0000 kWh, discharge = 134.8950 kWh, charge_state
 = 432.3200 kWh, price = \$0.0340
 (d=12.0, t=2.0) | charge = 205.0000 kWh, discharge = 133.6530 kWh, charge_state
 = 503.6670 kWh, price = \$0.0213
 (d=12.0, t=3.0) | charge = 205.0000 kWh, discharge = 135.1710 kWh, charge_state
 = 573.4960 kWh, price = \$0.0144
 (d=12.0, t=4.0) | charge = 205.0000 kWh, discharge = 135.4470 kWh, charge_state
 = 643.0490 kWh, price = \$0.0184
 (d=12.0, t=5.0) | charge = 205.0000 kWh, discharge = 132.5490 kWh, charge_state
 = 715.5000 kWh, price = \$0.0259
 (d=12.0, t=6.0) | charge = 180.0000 kWh, discharge = 194.9250 kWh, charge_state
 = 700.5750 kWh, price = \$0.0450
 (d=12.0, t=7.0) | charge = 160.0000 kWh, discharge = 239.1540 kWh, charge_state
 = 621.4210 kWh, price = \$0.0461
 (d=12.0, t=8.0) | charge = 160.0000 kWh, discharge = 231.6330 kWh, charge_state
 = 549.7880 kWh, price = \$0.0507
 (d=12.0, t=9.0) | charge = 160.0000 kWh, discharge = 236.9460 kWh, charge_state
 = 472.8420 kWh, price = \$0.0570
 (d=12.0, t=10.0) | charge = 150.0000 kWh, discharge = 256.8870 kWh, charge_state
 = 365.9550 kWh, price = \$0.0744
 (d=12.0, t=11.0) | charge = 150.0000 kWh, discharge = 255.5070 kWh, charge_state
 = 260.4480 kWh, price = \$0.0842
 (d=12.0, t=12.0) | charge = 134.1570 kWh, discharge = 251.5050 kWh, charge_state
 = 143.1000 kWh, price = \$0.0992
 (d=12.0, t=21.0) | charge = 101.7750 kWh, discharge = 101.7750 kWh, charge_state
 = 143.1000 kWh, price = \$0.0802
 (d=12.0, t=22.0) | charge = 100.4640 kWh, discharge = 100.4640 kWh, charge_state

= 143.1000 kWh, price = \$0.0717
 (d=12.0, t=23.0) | charge = 97.7040 kWh, discharge = 97.7040 kWh, charge_state =
 143.1000 kWh, price = \$0.0606
 (d=13.0, t=0.0) | charge = 97.0830 kWh, discharge = 97.0830 kWh, charge_state =
 143.1000 kWh, price = \$0.0577
 (d=13.0, t=1.0) | charge = 190.0000 kWh, discharge = 96.5310 kWh, charge_state =
 236.5690 kWh, price = \$0.0539
 (d=13.0, t=2.0) | charge = 220.0000 kWh, discharge = 93.2880 kWh, charge_state =
 363.2810 kWh, price = \$0.0484
 (d=13.0, t=3.0) | charge = 220.0000 kWh, discharge = 93.6330 kWh, charge_state =
 489.6480 kWh, price = \$0.0428
 (d=13.0, t=4.0) | charge = 220.0000 kWh, discharge = 92.1150 kWh, charge_state =
 617.5330 kWh, price = \$0.0411
 (d=13.0, t=5.0) | charge = 210.0000 kWh, discharge = 118.2660 kWh, charge_state =
 709.2670 kWh, price = \$0.0487
 (d=13.0, t=6.0) | charge = 181.3550 kWh, discharge = 175.1220 kWh, charge_state =
 715.5000 kWh, price = \$0.0577
 (d=13.0, t=7.0) | charge = 175.0000 kWh, discharge = 196.9260 kWh, charge_state =
 693.5740 kWh, price = \$0.0536
 (d=13.0, t=8.0) | charge = 175.0000 kWh, discharge = 196.9950 kWh, charge_state =
 671.5790 kWh, price = \$0.0598
 (d=13.0, t=9.0) | charge = 170.0000 kWh, discharge = 213.1410 kWh, charge_state =
 628.4380 kWh, price = \$0.0661
 (d=13.0, t=10.0) | charge = 87.9140 kWh, discharge = 223.7670 kWh, charge_state =
 492.5850 kWh, price = \$0.0788
 (d=13.0, t=11.0) | charge = 0.0000 kWh, discharge = 226.4580 kWh, charge_state =
 266.1270 kWh, price = \$0.0890
 (d=13.0, t=21.0) | charge = 0.0000 kWh, discharge = 123.0270 kWh, charge_state =
 143.1000 kWh, price = \$0.0853
 (d=13.0, t=22.0) | charge = 122.1990 kWh, discharge = 122.1990 kWh, charge_state =
 143.1000 kWh, price = \$0.0756
 (d=13.0, t=23.0) | charge = 142.7140 kWh, discharge = 121.9230 kWh, charge_state =
 163.8910 kWh, price = \$0.0669
 (d=14.0, t=0.0) | charge = 210.0000 kWh, discharge = 123.1650 kWh, charge_state =
 250.7260 kWh, price = \$0.0650
 (d=14.0, t=1.0) | charge = 210.0000 kWh, discharge = 123.3030 kWh, charge_state =
 337.4230 kWh, price = \$0.0586
 (d=14.0, t=2.0) | charge = 205.0000 kWh, discharge = 127.3050 kWh, charge_state =
 415.1180 kWh, price = \$0.0549
 (d=14.0, t=3.0) | charge = 215.0000 kWh, discharge = 109.7790 kWh, charge_state =
 520.3390 kWh, price = \$0.0532
 (d=14.0, t=4.0) | charge = 220.0000 kWh, discharge = 99.0840 kWh, charge_state =
 641.2550 kWh, price = \$0.0529
 (d=14.0, t=5.0) | charge = 205.0000 kWh, discharge = 130.7550 kWh, charge_state =
 715.5000 kWh, price = \$0.0575
 (d=14.0, t=6.0) | charge = 180.0000 kWh, discharge = 191.9580 kWh, charge_state =
 703.5420 kWh, price = \$0.0612
 (d=14.0, t=7.0) | charge = 185.0000 kWh, discharge = 177.0540 kWh, charge_state

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= 711.4880 kWh, price = $0.0639
(d=14.0, t=8.0) | charge = 180.0000 kWh, discharge = 192.2340 kWh, charge_state
= 699.2540 kWh, price = $0.0668
(d=14.0, t=9.0) | charge = 165.0000 kWh, discharge = 219.3510 kWh, charge_state
= 644.9030 kWh, price = $0.0696
(d=14.0, t=10.0) | charge = 160.0000 kWh, discharge = 236.1870 kWh, charge_state
= 568.7160 kWh, price = $0.0801
(d=14.0, t=11.0) | charge = 0.0000 kWh, discharge = 238.1190 kWh, charge_state =
330.5970 kWh, price = $0.0883
(d=14.0, t=21.0) | charge = 0.0000 kWh, discharge = 98.2560 kWh, charge_state =
232.3410 kWh, price = $0.0943
(d=14.0, t=22.0) | charge = 8.1180 kWh, discharge = 97.3590 kWh, charge_state =
143.1000 kWh, price = $0.0818
(d=14.0, t=23.0) | charge = 94.4610 kWh, discharge = 94.4610 kWh, charge_state =
143.1000 kWh, price = $0.0697
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